ME 5309 – Mid-Semester Examination: ANSWER SHEET

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Questions attempted: Q1, Q2, Q3

Question 1

Starting with this one because it looks the shortest.

Given: sensor 1280 px wide, pixel pitch p = 5 µm (square), lens f = 16 mm, standoff Z = 400 mm; force sensor 10 mV/N, 12-bit ADC over 0–5 V; plant $\dot{x} = -2x + u$ with zero-order hold, T = 0.1 s.

(a) (i) $f_x = f/p = 16\text{ mm} / 5\ \mu\text{m} =$ **3200 px**.

(ii) Sensor width = 1280 × 5 µm = 6.4 mm; FOV = $W_s \cdot Z/f = 6.4 \times 400/16 =$ **160 mm**.

(b) Assumptions: ideal 12-bit ADC (4096 codes), the full 0–5 V range is used, no offset or gain error, and sensor noise is below one LSB so quantisation sets the limit.

Voltage step: 1 LSB = $V_{FS}/2^N = 5\text{ V}/4096 = 1.2207 \times 10^{-3}$ V = 1.2207 mV.

The sensor maps force to voltage by V = S·F with S = 10 mV/N, so $\Delta F_{min} = \Delta V/S = 1.2207\text{ mV} / (10\text{ mV/N}) = 0.12207$ N. $\mathbf{\Delta F_{min} \approx 0.122\ N}$. (Using 4095 steps instead of 4096 codes gives 0.1221 N, the same at 3 s.f.)

(c) $F = e^{AT}$ and $G = \int_0^T e^{A\tau} B\, d\tau$.

$F = e^{-0.2} =$ **0.8187**.

$G = \int_0^{0.1} e^{-2\tau} d\tau = 1 - e^{-0.2} =$ **0.1813**.

Question 2

I think I know the first two parts, the third one is tricky.

Given: J = 0.02 kg·m², b = 0.05 N·m·s/rad, m = 1 kg, ℓ = 0.25 m, g = 9.81 m/s². So mgℓ = 1 × 9.81 × 0.25 = 2.4525 N·m (used throughout).

(a) $x_1 = \theta$, $x_2 = \dot{\theta}$. $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - b\, x_2 - mg\ell \sin x_1)/J$.

At equilibrium $x_2 = 0$ and $\sin\theta_e = \tau_0/(mg\ell)$, so $0 \le \tau_0 \le mg\ell = 2.45$ N·m.

(b) The equilibrium at $\theta_e = 60°$ needs the torque $\tau_e = mg\ell \sin 60° = 2.4525 \times 0.8660 = 2.124$ N·m. Define deviations $\delta x = x - x_e$ with $x_e = [60°, 0]^T$ and $\delta\tau = \tau - \tau_e$; then $\delta\dot{x} \approx A\,\delta x + B\,\delta\tau$ with the Jacobians evaluated at the equilibrium.

$\partial f_1/\partial x = [0, 1]$. $\partial f_2/\partial x_1 = -(mg\ell/J)\cos x_1 = -(2.4525/0.02)(0.5) = -122.625 \times 0.5 = -61.31$ (s^{-2}). $\partial f_2/\partial x_2 = -b/J = -0.05/0.02 = -2.5\ s^{-1}$.

A = [0, 1; −61.31, −2.5], and sence $\partial f_2/\partial\tau = 1/J = 50$, **B = [0; 50]**.

Check: the characteristic equation $\lambda^2 + 2.5\lambda + 61.31 = 0$ gives $\lambda = -1.25 \pm 7.73j$, a lightly damped stable oscillation, as expected for a pendulum hanging near-vertical (60° below the downward vertical).

(c) Set all time derivatives to zero in the closed loop. With $\tau = K_p(\theta_d - \theta) - K_d\dot{\theta}$ and $\dot{\theta} = \ddot{\theta} = 0$ the damper term vanishes (so K_d has no effect on the final position) and the dynamics reduce to $mg\ell \sin\theta_{ss} = K_p(\theta_d - \theta_{ss})$.

Why $\theta_{ss} \neq \theta_d$: if the joint were at rest exactly at $\theta_d = 90°$ the PD torqe would be $K_p \cdot 0 - K_d \cdot 0 = 0$, yet gravity pulls with $mg\ell \sin 90° = 2.4525$ N·m. The net torqe would be non-zero, so the joint could not be at rest. A purely proportional-derivative law can only hold a load if there is a stedy error.

Let $e = \theta_d - \theta_{ss}$ (in rad). Then $\sin\theta_{ss} = \sin(\pi/2 - e) = \cos e$, and the equation becomes $50\, e = 2.4525 \cos e$. Fixed-point iteration: $e_0 = 2.4525/50 = 0.04905$ (taking $\cos e \approx 1$); $e_1 = 0.04905 \times \cos(0.04905) = 0.04899$; $e_2 = 0.04905 \times \cos(0.04899) = 0.04899$, converged.

e = 0.04899 rad × (180/π) = 2.807°, so θ_{ss} = 87.19°. **Steady-state error ≈ 2.81° (the arm sags below the target).** This is small becuase K_p is much larger than mgℓ (K_p/mgℓ ≈ 20).

(d) Gravity compensation reduces the stedy-state error to about 0.5°, not exactly zero, since the compensation is evaluated at the measured angel and the sensor and model are not perfect.

Poles: unchanged. The linearisation of the graviy term about θ_d is mgℓ cos θ_d, wich is zero at 90°, so the stiffness stays K_p.

Skipping the check, running out of time. Hope partial marks are given for method.

Question 3

Observer question. I remember the idea from class, hope I get the algebra right.

Given: A = [0, 1; 0, −2], B = [0; 1], C = [1, 0], f_s = 1 kHz so the sample time is T = 1/f_s = 1 ms.

(a) Reason 1 – noise amplification: the backward difference $\hat{\omega}_k = (\theta_k - \theta_{k-1})/T$ has frequency response $(1 - e^{-j\omega T})/T$, whose magnitude grows with frequency (≈ ω for small ωT), i.e. it is a high-pass operation. Any noise on the encoder signal therefore appears in the velocity multiplied by roughly 1/T = 1000 s^{-1}, and can swamp the true velocity.

Reason 2 – quantisaton: the encoder only reports integer counts, so Δθ is an integer multiple of the count size q. Dividing by T = 1 ms gives velocity steps of q/T; for a 4096-count encoder q = 2π/4096 = 1.53×10^{-3} rad, so one step is 1.53 rad/s. At low speed the estimate flips between 0 and one step (a noisy staircase), exactly where the damping term of the PD law is needed.

(A further, smaller issue is the half-sample delay of a backward difference, but the two reasons above are the main ones.)

(b) The analogue prototype is the first-order low-pass H(s) = $\omega_c/(s + \omega_c)$ with $\omega_c = 2\pi f_c = 2\pi \times 20$ = 125.66 rad/s. The bilinear (Tustin) transform replaces s by (2/T)(z − 1)/(z + 1) with T = 1/f_s = 0.001 s.

Define a = ω_cT/2 = 125.66 × 0.001/2 = 0.062832. Then H(z) = ω_c / [(2/T)(z−1)/(z+1) + ω_c] = a(z + 1) / [(1 + a) z − (1 − a)] = a(1 + z^{-1}) / [(1 + a) − (1 − a) z^{-1}].

Dividing through by (1 + a): a/(1 + a) = 0.062832/1.062832 = 0.059117 and (1 − a)/(1 + a) = 0.937168/1.062832 = 0.881767. Hence with u = differenced signal and y = filtered velocity: $\mathbf{y_k = 0.05912\,(u_k + u_{k-1}) + 0.88177\,y_{k-1}}$. (Frequency pre-warping is neglected since $f_c \ll f_s/2$.)

Verification of the DC gain: set u and y constant (z = 1): y = 2(0.05912)u + 0.88177 y ⇒ y(1 − 0.88177) = 0.11824 u ⇒ y/u = 0.11824/0.11823 = 1.000. In general H(1) = 2a/[(1 + a) − (1 − a)] = 2a/2a = 1 exactly, so the stedy-state output equals the commanded input ✓.

(c) A − LC = [$-\ell_1$, 1; $-\ell_2$, −2]; characteristic polynomial det(sI − A + LC) = (s + ℓ_1)(s + 2) + ℓ_2 = s^2 + (2 + ℓ_1)s + (2ℓ_1 + ℓ_2).

Desired $(s + 10)^2 = s^2 + 20s + 100$. Then 2 + ℓ_1 = 20 ⇒ ℓ_1 = 18, and 2ℓ_1 + ℓ_2 = 100 ⇒ ℓ_2 = 100 − 18 = 82.

L = [18, 82]^T.

(d) An observer works only when there is no noice, so it has no advantage; a filter is always better becuase it is simpler.

The filter is safer when the model is perfectly known.